Computer Networks

Complete Notes — Basic to Advanced

Placement & Interview Ready | Easy Language | Examples | Diagrams

Abhishek Kashyap

Sharing Knowledge · Helping Everyone Crack Placements

Network Basics | OSI 7 Layers | TCP/IP | Protocols | IP Addressing | DNS & HTTP | Subnetting | Security | Interview Q&A

Created for Placement & Interview Prep · Easy Language · Diagrams · Interview Tips

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1 What is a Computer Network?

A **computer network** is a group of two or more devices (computers, phones, printers, servers) connected together so they can share data and resources. Think of it like a group WhatsApp — everyone is connected, anyone can share files, and messages reach everyone instantly.

Networks are everywhere! When you browse the internet, stream Netflix, send an email, or print a document wirelessly — a network is working behind the scenes every single time.

Key Networking Terms

Term	Meaning
Node	Any device on a network — laptop, phone, printer, router, server
Link	Medium connecting two nodes — Ethernet cable, Wi-Fi signal, fiber optic
Bandwidth	Maximum data transfer speed a link can support (e.g. 1 Gbps Ethernet)
Throughput	Actual data speed achieved in practice — always ≤ bandwidth
Latency	Delay for data to travel from source to destination (measured in ms)
Protocol	Agreed set of rules for communication — like a common language both sides follow
Packet	Data broken into small chunks for sending; reassembled at destination
Port	Number (0–65535) identifying which application data belongs to on a device
IP Address	Unique logical address of a device on a network — like a postal address
MAC Address	Unique physical address burned into NIC by manufacturer — permanent fingerprint

Real-Life Example

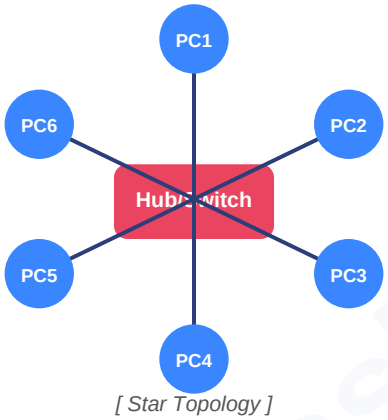
Your home Wi-Fi is a network! Router connects phone+laptop+TV (LAN).

Router then connects home network to the internet via ISP (WAN connection).

2 Network Topologies

Topology is the **physical or logical layout** of a network — how devices are arranged and connected. The right topology affects cost, reliability, and performance.

1. Star Topology



[Star Topology]

Every device connects **directly to one central hub or switch**. No device communicates directly with another — all traffic goes through the center. Like a wheel — hub is the center, devices are the spokes.

Advantages:

- Easy to add/remove devices — just plug into the central switch
- One cable failing only disconnects that one device; others unaffected
- Easy to troubleshoot — isolate faults quickly

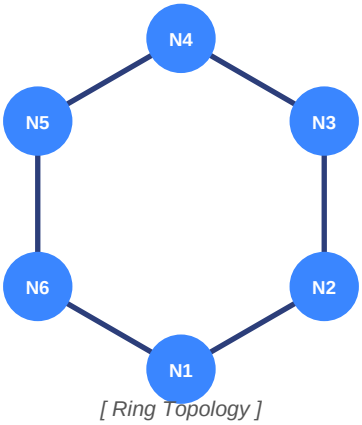
Disadvantages:

- Central hub/switch failure brings down the ENTIRE network
- Requires more cable than Bus — one dedicated cable per device

Used In

Office LANs, School labs, Home Wi-Fi — MOST widely used topology today!

2. Ring Topology



[Ring Topology]

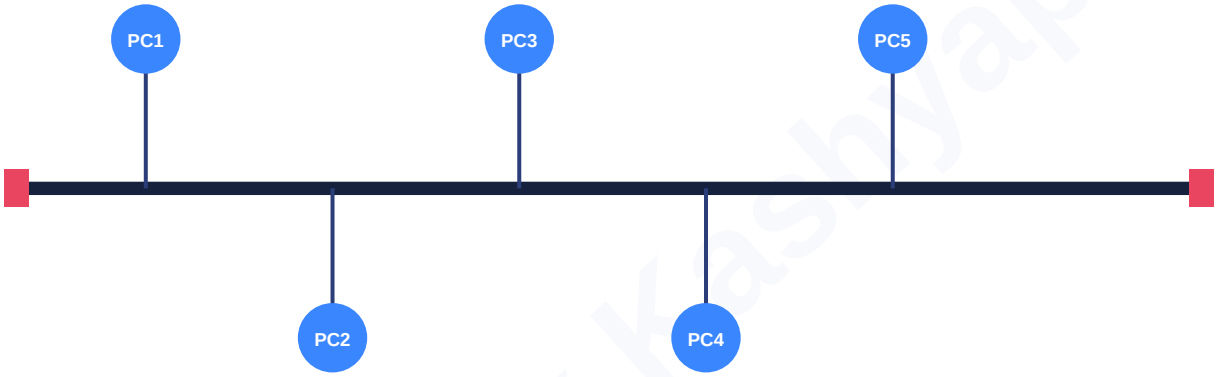
Each device connects to **exactly two neighbours**, forming a closed loop. Data travels around the ring in one direction, passing through each node until it reaches its destination.

- ■ No central device needed — equal access for all nodes
- ■ Performs well under heavy load — predictable token passing
- ■ Single node or cable failure breaks the entire ring
- ■ Expensive, hard to reconfigure; adding a node requires ring break

Example

SONET/SDH telecom backbone. Token Ring (IEEE 802.5) — now replaced by Ethernet.

3. Bus Topology



[Bus Topology]

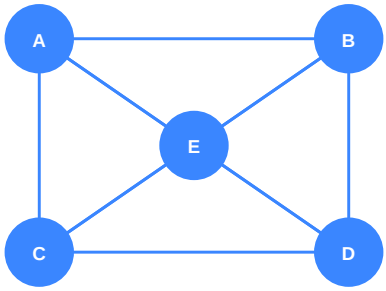
All devices share **one common backbone cable (the Bus)**. Data travels the entire cable; each device checks if data is for it — like a PA announcement on a school corridor.

- ■ Cheapest and simplest — minimal cable required
- ■ Bus cable failure → entire network down
- ■ Performance drops with more devices; collisions possible (uses CSMA/CD)

Note

Early Ethernet (10BASE2, 10BASE5) used Bus. Rarely used today — replaced by Star.

4. Mesh Topology



[Mesh Topology]

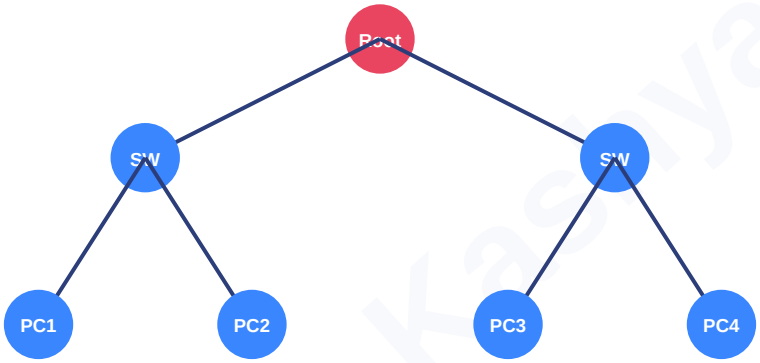
Every device has a **dedicated point-to-point link to every other device**. Multiple redundant paths exist — if one fails, data takes another. Like a spider web.

- Full Mesh: every node ↔ every node. Cables needed = n(n–1)/2
- Partial Mesh: only some nodes directly connected — reduces cost
- ■ Most reliable — single link failure has no impact
- ■ Expensive — large cabling cost; complex management

Formula

5 nodes → 5×4/2 = 10 cables. Used in: Military, Telecom core, Internet backbone.

5. Tree Topology



[Tree Topology]

A **Star + Bus hybrid**. Multiple star networks hang off a common bus backbone — like a tree trunk with branches. Each branch (star segment) works independently.

- ■ Hierarchical — easy to manage and expand
- ■ Segment failure isolated — other branches unaffected
- ■ Backbone/bus failure brings down everything

Used In

Multi-floor offices, University campuses — each floor has a star, connected to central backbone.

6. Hybrid Topology

Combination of two or more topology types. Most real enterprise networks are hybrid because different departments may need different topologies.

- Star + Ring = Hybrid (connect multiple star clusters in a ring)
- Star + Star = Still Star (same type = no hybrid)
- ■ Flexible and scalable; ■ Complex and costly to design

3 Types of Networks — PAN, LAN, MAN, WAN, GAN

Networks classified by coverage area — from your pocket to the entire planet!

Type	Full Form	Range	Speed	Example
PAN	Personal Area Network	~10 m	High	Bluetooth, USB tethering, smartwatch
HAN	House Area Network	Within house	High	Home Wi-Fi (phone+laptop+TV)
LAN	Local Area Network	Building/Campus	1 Gbps+	Office, School, Cyber cafe
CAN	Campus Area Network	Multiple buildings	High	University, IT Park, Hospital
MAN	Metropolitan Area Network	City-wide	High	Cable TV, City Wi-Fi network
WAN	Wide Area Network	Countries/Continents	Variable	Internet, Bank ATM network
GAN	Global Area Network	Worldwide	Variable	Satellite internet, Undersea cables

Memory Trick

"Please Help Local Campus Make Wide Global Networks"

P=PAN, H=HAN, L=LAN, C=CAN, M=MAN, W=WAN, G=GAN — Smallest to Largest!

VPN — Virtual Private Network

VPN creates an **encrypted secure tunnel over the public internet**. Like whispering through a soundproof pipe in a crowded room — only you and the receiver hear it.

VPN Type	Who Uses It	How It Works
Access VPN	Individual remote workers	Employee connects from home to office server securely
Site-to-Site	Offices in different cities	Mumbai ↔ Delhi office — work as if in same building
Intranet VPN	Departments of same company	Different branches share resources securely
Extranet VPN	Partners, suppliers, clients	External parties get limited controlled access

Interview Tip

VPN protocols: IPSec, L2TP, PPTP, OpenVPN, WireGuard, SSL VPN.

VPN hides real IP, encrypts traffic, and lets you bypass geo-restrictions safely.

4 IP Addressing — IPv4, Classes, Private/Public & IPv6

Every device needs a unique address so data reaches the right place. This is the **IP (Internet Protocol) Address** — like your home postal address on the internet.

IPv4 — Structure

IPv4 is a **32-bit address** in "dotted decimal" — 4 octets of 8 bits each, separated by dots. Each octet: 0–255. Two parts: Network portion (identifies the network) + Host portion (the device).

Example

192.168.1.100 → 4 octets → 32 bits total

11000000.10101000.00000001.01100100 (binary)

IPv4 Classes

Class	First Octet	Subnet Mask	Net Bits	Max Hosts	Used For
A	1 – 126	255.0.0.0	8	16,777,214	Very large networks (ISPs, govts)
B	128 – 191	255.255.0.0	16	65,534	Medium networks (universities)
C	192 – 223	255.255.255.0	24	254	Small networks — homes & offices
D	224 – 239	N/A	N/A	N/A	Multicasting only
E	240 – 255	N/A	N/A	N/A	Experimental & research

Special Addresses

127.0.0.1 = Loopback / localhost (your own machine — used for testing)

0.0.0.0 = Unspecified / this network

255.255.255.255 = Limited broadcast (all hosts on local network)

Private vs Public IP

Private IPs are used inside home/office — NOT routable on internet. Router assigns them via DHCP. Three reserved ranges:

Range	Class	Example Use
10.0.0.0 – 10.255.255.255	A	Large enterprise internal network
172.16.0.0 – 172.31.255.255	B	Medium company internal network
192.168.0.0 – 192.168.255.255	C	Home network (192.168.1.x most common)

Public IPs are globally unique, assigned by ISP, visible on internet. Your router has one public IP representing ALL home devices to the outside world.

NAT — Network Address Translation

NAT lets many devices share ONE public IP.

Router translates private IP → public IP when sending; reverses on receiving.

Example: All 6 devices in your home share ISP public IP 103.28.14.22.

IPv6 — Next Generation

IPv4 supports ~4.3 billion addresses — we have run out! **IPv6 = 128-bit address** → 3.4×10^{3}■ addresses. Written as 8 groups of 16 bits in hexadecimal separated by colons.

IPv6 Example

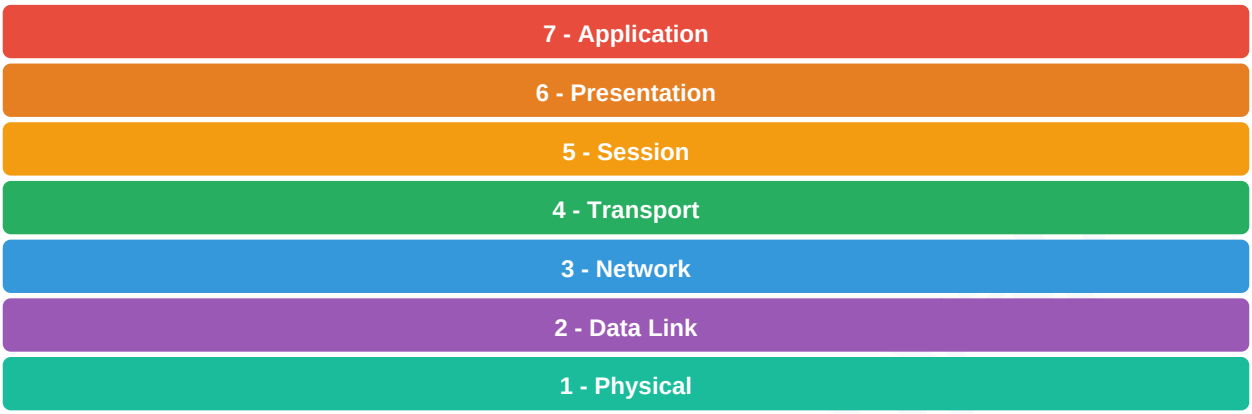
2001:0db8:85a3:0000:0000:8a2e:0370:7334

Improvements: No NAT needed · Built-in IPSec · Auto-configuration · Better routing

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5 OSI Model — All 7 Layers Explained

OSI (Open Systems Interconnection) is a **7-layer reference model** describing how data travels across a network. Developed by ISO. Each layer has a specific, well-defined job — like floors of a building, each doing its own work.



[OSI Model (7 Layers)]

Memory Trick — Layer 7 down to 1

"All People Seem To Need Data Processing"

A=Application, P=Presentation, S=Session, T=Transport, N=Network, D=DataLink, P=Physical

Layer 7: Application Layer

Protocols: HTTP, HTTPS, FTP, SMTP, DNS, POP3, SSH

The layer YOU interact with. Provides network services to user applications (Chrome, Gmail, WhatsApp). Not the app itself — the interface that lets apps use network services.

Example: Browser fetching a webpage; Gmail sending an email

Layer 6: Presentation Layer

Protocols: SSL/TLS, JPEG, MP4, ASCII, Unicode, MPEG

The "translator". Converts data to a standard format, handles ENCRYPTION (scrambles data for security) and COMPRESSION (reduces data size). Ensures sender and receiver understand each other.

Example: SSL encrypting your password before sending it

Layer 5: Session Layer

Protocols: NetBIOS, PPTP, RPC, NFS, SIP

Manages the full lifecycle of a communication: ESTABLISHMENT, MAINTENANCE, and TERMINATION of sessions. Also handles synchronization checkpoints — so large transfers can resume after interruption.

Example: A video call being set up and disconnected; login sessions on websites

Layer 4: Transport Layer

Protocols: TCP (reliable), UDP (fast)

Responsible for END-TO-END delivery. Breaks data into SEGMENTS (TCP) or DATAGRAMS (UDP). Uses PORT NUMBERS to identify applications. TCP guarantees delivery; UDP prioritises speed.

Example: TCP ensuring all packets of a file download arrive in correct order

Layer 3: Network Layer

Protocols: IP (v4/v6), ICMP, ARP, OSPF, BGP, RIP

Handles LOGICAL ADDRESSING (IP addresses) and ROUTING — finding the best path across networks. Breaks data into PACKETS. Routers operate at this layer.

Example: Router choosing the best path from India to a server in the USA

Layer 2: Data Link Layer

Protocols: Ethernet, Wi-Fi (802.11), MAC, PPP, ARP

Handles PHYSICAL ADDRESSING (MAC addresses), FRAMING (wraps packets into frames), and ERROR DETECTION between two directly connected nodes. Two sublayers: LLC and MAC. Switches operate here.

Example: Switch forwarding a frame to correct device based on its MAC address

Layer 1: Physical Layer

Protocols: Ethernet cable, Fiber optic, Wi-Fi radio, USB

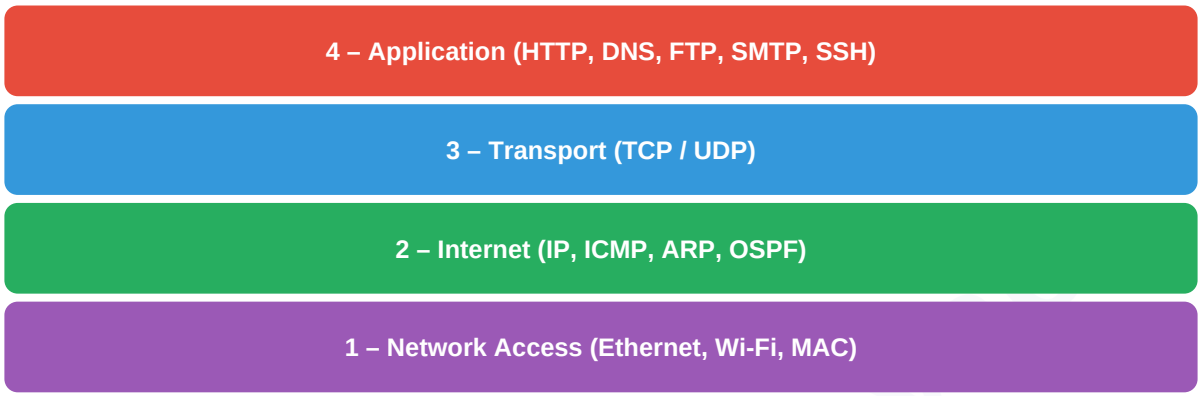
Actual physical transmission of raw BITS (0s and 1s). Defines electrical signals, radio waves, light pulses, connectors, and data rates. No understanding of frames or packets — just bits.

Example: Electrical signals traveling through an Ethernet cable between two computers

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6 TCP/IP Model & 3-Way Handshake

TCP/IP is the **actual protocol suite the real internet runs on!** Developed by DARPA (US Dept of Defense). It has 4 layers — combines OSI's top 3 layers into one Application layer.

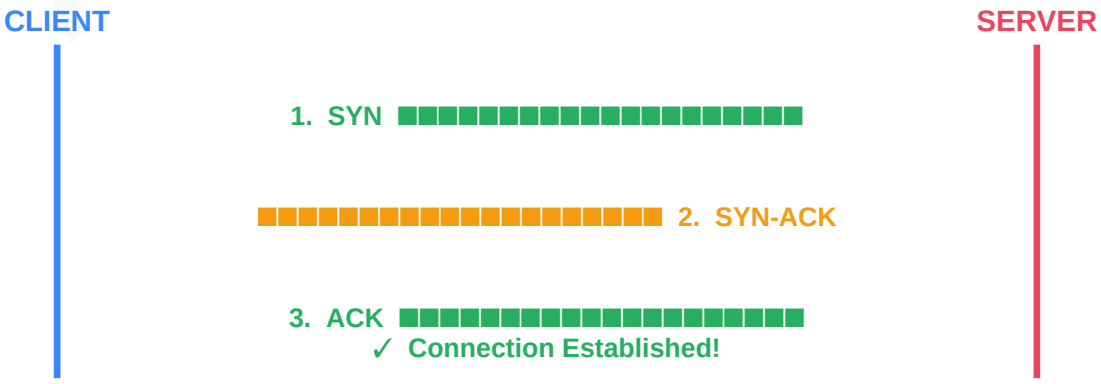


[TCP/IP Model (4 Layers)]

TCP/IP Layer	Equivalent OSI	Protocols	Function
Application (4)	OSI 5+6+7	HTTP, HTTPS, FTP, SMTP, DNS, SSH	User-facing services
Transport (3)	OSI 4	TCP, UDP	End-to-end delivery, port numbers
Internet (2)	OSI 3	IP, ICMP, ARP, OSPF, BGP	Logical addressing & routing
Network Access (1)	OSI 1+2	Ethernet, Wi-Fi, MAC	Physical transmission & framing

TCP 3-Way Handshake — Connection Setup

Before TCP sends any data, a connection is established via 3 steps. Like calling someone: ring (SYN) → answer (SYN-ACK) → start talking (ACK).



[TCP 3-Way Handshake]

- **Step 1 — SYN:** Client sends SYN packet. "I want to connect. My sequence starts at X."
- **Step 2 — SYN-ACK:** Server replies. "Got your SYN (ACK=X+1). My sequence starts at Y."
- **Step 3 — ACK:** Client confirms. "Got your SYN-ACK (ACK=Y+1). We are connected!"

4-Way Connection Termination

To CLOSE a TCP connection: Client FIN → Server ACK → Server FIN → Client ACK

Both sides must independently close — that is why it needs 4 steps instead of 3.

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7 TCP vs UDP — Detailed Comparison

Two main Transport Layer protocols. Know the differences cold for interviews!

Feature	TCP	UDP
Full Form	Transmission Control Protocol	User Datagram Protocol
Connection	Connection-oriented (3-way handshake)	Connectionless (just send)
Reliability	■ Guaranteed delivery + retransmission	■ No guarantee — lost = lost
Ordering	■ Correct order maintained	■ May arrive out of order
Speed	■ Slower (ACK overhead)	■ Much faster (no overhead)
Header	20 bytes minimum	8 bytes only
Flow Control	■ Sliding window	■ None
Error Check	Checksum + ACK + retransmit	Checksum only
Use Cases	Web, Email, File Transfer, SSH	Video calls, Gaming, DNS, Streaming
Protocols	HTTP, HTTPS, FTP, SMTP, SSH	DNS, DHCP, VoIP, YouTube Live

Simple Analogy

TCP = Registered post: tracked, confirmed, resent if lost.

UDP = Throwing a flyer from a window: fast, no confirmation it was received!

8 Important Protocols & Port Numbers

A **protocol** is a standardised set of rules. A **port number** identifies the application on a device. IP address = building; port = flat number inside.

Protocol	Full Name	Port	Layer	Purpose
HTTP	HyperText Transfer Protocol	80	App	Web pages — plain, unencrypted
HTTPS	HTTP Secure	443	App	Secure web browsing (TLS encrypted)
FTP	File Transfer Protocol	21/20	App	Transfer files between hosts
SFTP	SSH File Transfer Protocol	22	App	Secure encrypted file transfer
SSH	Secure Shell	22	App	Secure remote login & commands
Telnet	Telnet	23	App	Remote login — NOT secure (plain text)
SMTP	Simple Mail Transfer Protocol	25	App	Sending emails between servers
POP3	Post Office Protocol v3	110	App	Download emails to device
IMAP	Internet Message Access Protocol	143	App	Access emails on server
DNS	Domain Name System	53	App	Domain → IP translation
DHCP	Dynamic Host Config Protocol	67/68	App	Auto-assign IPs to devices
SNMP	Simple Network Mgmt Protocol	161	App	Monitor/manage network devices
ICMP	Internet Control Message Protocol	—	Net	Error reporting, Ping
ARP	Address Resolution Protocol	—	Net	IP address → MAC address
RIP	Routing Information Protocol	520	Net	Distance-vector routing
OSPF	Open Shortest Path First	89	Net	Link-state routing (Dijkstra)
BGP	Border Gateway Protocol	179	Net	Inter-ISP internet routing

9 DNS — Domain Name System (Deep Dive)

DNS is the **phonebook of the internet**. Instead of memorising "142.250.195.14" you type "google.com" and DNS translates it. Invented by Paul Mockapetris & Jon Postel (1983).

DNS Resolution — Step by Step

1. Browser Cache

Browser checks its own DNS cache. If found and still valid (within TTL) → use it directly. No network needed!

2. OS Cache

Browser asks OS. OS checks its cache and the local hosts file (/etc/hosts on Linux, C:\Windows\System32\drivers\etc\hosts on Windows).

3. Recursive Resolver

OS queries your ISP's DNS Recursive Resolver. This server does all the legwork on your behalf.

4. Root DNS Server

Resolver asks one of 13 Root Server clusters (worldwide). Root doesn't know the IP but says: "For .com, ask this TLD server."

5. TLD Name Server

Resolver asks the .com TLD server. It says: "For google.com, ask Google's Authoritative server at this address."

6. Authoritative DNS

Google's own DNS server — final authority for google.com — returns the actual IP: 142.250.195.14

7. Response & Cache

Resolver returns IP → OS tells browser → browser connects. Result cached at each level for TTL duration.

DNS Record Types

A Record: domain → IPv4 (google.com → 142.250.195.14)

AAAA Record: domain → IPv6

CNAME: Alias (www.google.com → google.com)

MX Record: Mail servers for a domain

NS Record: Authoritative name servers

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10 HTTP vs HTTPS & Network Security

HTTP vs HTTPS Comparison

Feature	HTTP	HTTPS
Port	80	443
Security	No encryption — plain text	SSL/TLS encrypted — secure
Data Safety	Anyone can intercept and read	Unreadable even if intercepted
Certificate	Not required	SSL/TLS cert from Certificate Authority
Browser	No padlock ■	Padlock ■ in address bar
SEO	Lower Google ranking	Higher ranking — Google prefers HTTPS
Use Case	Old/internal sites	ALL modern sites — banking, shopping, etc.

SSL/TLS Handshake — How HTTPS Works

Before any data, browser and server perform TLS handshake to establish encryption:

- **Client Hello:** Browser sends supported TLS versions and cipher suites
- **Server Hello:** Server picks best cipher; sends its SSL Certificate (contains public key)
- **Certificate Verify:** Browser verifies cert with trusted CA (DigiCert, Let's Encrypt, etc.)
- **Key Exchange:** Browser generates session key, encrypts with server's public key, sends it
- **Secure Channel:** Both use shared session key for fast symmetric encryption (AES) going forward

Symmetric vs Asymmetric

Asymmetric (RSA): public+private key pair — slow; used only for handshake.

Symmetric (AES): single shared key — fast; used for actual data encryption.

HTTPS uses BOTH: asymmetric to exchange key safely, symmetric for all data.

Common Network Attacks & Prevention

Attack	What Happens	Prevention
DDoS	Flood of fake requests crashes the server	Rate limiting, CDN, Firewall, Traffic scrubbing
Man-in-Middle	Attacker sits between client/server, reads/modifies data	HTTPS/TLS, Certificate pinning, VPN
DNS Spoofing	Fake DNS response redirects user to malicious site	DNSSEC, Trusted DNS (8.8.8.8), HTTPS
ARP Poisoning	Fake ARP reply links attacker MAC to victim IP	Static ARP, VLANs, IDS/IPS
Phishing	Fake site tricks user into entering credentials	User awareness, MFA, Email filters

Attack	What Happens	Prevention
Packet Sniffing	Attacker reads unencrypted network packets	Encrypt everything — HTTPS, VPN, SSH

Firewall

Firewall monitors ALL traffic and blocks/allows based on security rules.

Types: Packet Filter (IP/port rules) | Stateful (tracks connections) | Application/WAF (deep inspection)

Can be hardware device, software program, or cloud-based service.

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11 What Happens When You Type google.com? ■ MOST ASKED

The single MOST popular networking interview question. Know every step. Let's trace from keystroke to rendered page:

STEP 1	**Browser Cache**	Browser checks: "Did I visit recently, is cache fresh?" If yes → show instantly. No network call!
STEP 2	**DNS Resolution**	Browser cache → OS cache → Hosts file → ISP Resolver → Root DNS → .com TLD → Google Auth DNS. Returns IP: 142.250.195.14
STEP 3	**TCP Connection**	3-Way Handshake with Google server at port 443. SYN → SYN-ACK → ACK. Route: NIC → router → ISP → internet → Google data center.
STEP 4	**TLS Handshake**	HTTPS requires TLS: cert exchange → CA verification → session key negotiation → encrypted channel established.
STEP 5	**HTTP GET Request**	Browser sends: GET / HTTP/1.1 | Host: www.google.com | with headers (User-Agent, Cookies, Accept-Language, etc.)
STEP 6	**Server Processing**	Load balancer receives request → routes to least-busy server → app logic runs → database/cache queried → HTML generated.
STEP 7	**HTTP Response**	Server sends: 200 OK | Content-Type: text/html | gzip-compressed HTML with headers.
STEP 8	**Browser Renders**	Browser parses HTML → builds DOM → downloads CSS/JS/images → builds Render tree → Layout → Paint → Display!
STEP 9	**Cache & Close**	Browser caches assets per Cache-Control headers. TCP kept alive (Keep-Alive) or closed with 4-way FIN handshake.

12 Subnetting & CIDR — With Worked Examples

Subnetting = **divide one large network into smaller subnets**. Benefits: better security, efficient routing, organized IP management. CIDR (Classless Inter-Domain Routing) uses prefix notation like /24 instead of old class-based masks.

CIDR Quick Reference Table

CIDR	Subnet Mask	Net Bits	Host Bits	Usable Hosts	Typical Use
/8	255.0.0.0	8	24	16,777,214	ISP backbone
/16	255.255.0.0	16	16	65,534	Large organization
/24	255.255.255.0	24	8	254	Typical office/home
/25	255.255.255.128	25	7	126	Half of /24
/26	255.255.255.192	26	6	62	Quarter of /24
/27	255.255.255.224	27	5	30	Small team
/28	255.255.255.240	28	4	14	Very small subnet
/30	255.255.255.252	30	2	2	Point-to-point link
/32	255.255.255.255	32	0	1 (host)	Single host route

Worked Example — /26 Subnetting

Question: Divide 192.168.10.0/24 into /26 subnets. How many subnets? Hosts each?

- Original: 192.168.10.0/24 (Class C, 24 network bits)
- New prefix /26 → borrowed bits = 26 – 24 = 2 bits
- Number of subnets = 2^2 = 4 subnets
- Host bits remaining = 32 – 26 = 6 bits
- Total addresses per subnet = 2■ = 64
- Usable hosts = 64 – 2 = 62 (–1 Network addr, –1 Broadcast addr)

Subnet	Network Address	Subnet Mask	Host Range	Broadcast	Hosts
1 (00)	192.168.10.0	/26	192.168.10.1 – .62	192.168.10.63	62
2 (01)	192.168.10.64	/26	192.168.10.65 – .126	192.168.10.127	62
3 (10)	192.168.10.128	/26	192.168.10.129 – .190	192.168.10.191	62
4 (11)	192.168.10.192	/26	192.168.10.193 – .254	192.168.10.255	62

Quick Formulas

Subnets = 2^(borrowed bits) | Total addresses = 2^(host bits) | Usable = 2^(host bits) – 2

First address = Network ID (unusable) | Last address = Broadcast (unusable)

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13 Top 15 Interview Questions & Answers

Most frequently asked CN questions in TCS, Infosys, Wipro, Amazon, Google, Microsoft placements. Understand every answer — don't just memorise!

Q1. OSI model vs TCP/IP model — what is the difference?

➤ OSI = 7-layer theoretical model for understanding (Physical, Data Link, Network, Transport, Session, Presentation, Application). TCP/IP = 4-layer real-world model the internet actually uses (Network Access, Internet, Transport, Application). TCP/IP merges top 3 OSI layers into one. OSI is a reference standard; TCP/IP is what runs the internet.

Q2. Difference between Hub, Switch and Router?

➤ Hub (Layer 1): Dumb — broadcasts to ALL ports, causes collisions, no intelligence. Switch (Layer 2): Smart — learns MAC addresses, sends data only to correct port, full-duplex, efficient. Router (Layer 3): Connects different networks using IP addresses, runs routing protocols (OSPF/BGP), has its own IP, finds best path between networks.

Q3. What is TCP 3-way handshake and why is it needed?

➤ Three steps to establish TCP connection: SYN (client wants to connect) → SYN-ACK (server agrees, acknowledges) → ACK (client confirms). Needed to: synchronize sequence numbers, ensure both ends are ready, agree on parameters before data flows. To close: 4-way FIN → ACK → FIN → ACK.

Q4. When would you use UDP instead of TCP?

➤ Use UDP when speed matters more than reliability: live video/audio streaming (small delay is OK), online gaming (latest position beats guaranteed delivery), DNS queries (small single packet — retransmit if no reply), DHCP, VoIP. App handles its own error correction, or losing a few packets is acceptable.

Q5. What is subnetting? What does /24 mean?

➤ Subnetting divides a large network into smaller subnets for better security, routing, and management. /24 (CIDR) means first 24 bits are the network part → subnet mask 255.255.255.0 → 32−24=8 host bits → 2■−2 = 254 usable hosts. /26 → 62 hosts. /30 → 2 hosts (point-to-point link).

Q6. Explain DNS and how it resolves a domain name.

➤ DNS = phonebook of internet. Maps domain names to IPs. Resolution order: browser cache → OS cache → ISP recursive resolver → Root DNS → TLD server (.com) → Authoritative DNS → IP returned. Uses UDP port 53 (TCP for large responses). Result cached per TTL value.

Q7. What is ARP and when is it used?

➤ ARP (Address Resolution Protocol) maps a known IP address to an unknown MAC address within a LAN. Process: device broadcasts "Who has IP x.x.x.x? Tell me your MAC!" → owner replies with its MAC → ARP cache stores mapping for future use. Needed because switches use MAC, but upper layers use IP.

Q8. What is NAT and why is it used?

➤ NAT (Network Address Translation) maps private IPs to one public IP. Your home router performs NAT — all home devices share one ISP-assigned public IP. Reasons: IPv4 shortage, hides internal IPs (security), saves cost. Types: Static NAT (1:1), Dynamic (pool), PAT/Masquerade (many:1 — most common).

Q9. What is the difference between MAC and IP address?

➤ MAC: 48-bit physical address, hardcoded by NIC manufacturer, format AA:BB:CC:DD:EE:FF, Data Link layer, local network only, permanent. IP: logical address, assigned by network/DHCP, global or local, Network layer, can change. MAC = Aadhaar number (permanent). IP = home address (can change when you move).

Q10. What does ping do and what is TTL?

➤ Ping sends ICMP Echo Request to target IP → target sends ICMP Echo Reply → ping measures Round Trip Time. Shows: packets sent/received/lost and min/max/avg RTT. TTL (Time To Live): counter in each packet that decrements by 1 at each router. When TTL hits 0, packet is discarded and ICMP "Time Exceeded" sent back. Prevents infinite loops.

Q11. How does DHCP assign IP addresses?

➤ DHCP uses DORA: Discover (client broadcasts "I need an IP!") → Offer (DHCP server offers IP, mask, gateway, DNS) → Request (client formally requests that IP) → Acknowledge (server confirms, lease starts). Server = usually your router. Uses UDP 67 (server) and 68 (client). Lease expires and renews automatically.

Q12. What is a Firewall? Types?

➤ Firewall monitors traffic and allows/blocks based on rules. Types: (1) Packet Filter — checks IP/port/protocol, stateless, fast but basic. (2) Stateful Inspection — tracks connection state, knows if packet belongs to established session. (3) Application/WAF — inspects content, detects attacks like SQL injection. (4) Next-Gen — combines all + IDS/IPS + deep packet inspection.

Q13. Difference between Unicast, Multicast, Broadcast, Anycast?

➤ Unicast: one sender → one specific receiver (HTTP request). Multicast: one → group of subscribers (live streaming, OSPF updates). Broadcast: one → ALL devices on network (DHCP Discover, ARP Request). Anycast: one → nearest member of a group (DNS root servers, CDN edge nodes — fastest one responds).

Q14. What is a Proxy? Forward proxy vs Reverse proxy?

➤ Forward Proxy: represents CLIENT to internet. Client → Proxy → Internet. Used for anonymity, content filtering, caching. Corporate networks use it to monitor/control internet access. Reverse Proxy: represents SERVER to clients. Internet → Proxy → Servers. Used for load balancing, SSL termination, caching, DDoS protection. Examples: Nginx, HAProxy, Cloudflare.

Q15. What is Bandwidth vs Latency vs Throughput?

➤ Bandwidth: max theoretical capacity (like highway width — number of lanes). Latency: time for one packet to travel source→destination (distance/travel time). Throughput: actual data delivered per second in practice (always ≤ bandwidth). Example: Satellite internet has high bandwidth (50 Mbps) but high latency (600 ms delay) — bad for gaming, OK for streaming.

All the Best to You All! ■ — Abhishek Kashyap

These notes cover everything you need: Basic Networks → OSI → TCP/IP → Protocols → IP Addressing → DNS → Security → Subnetting → Interview Q&A

Revise daily: OSI layers, TCP handshake, DNS resolution steps, subnetting formulas and port numbers.

Practice subnetting calculations on paper — they frequently appear in placement written rounds!

Hope these notes help you crack your dream company. Share with your friends too! ■

Abhishek Kashyap